

$$UCB := \frac{Q_c}{N_c} + C \sqrt{\frac{\ln N_{parent}}{N_c}}$$

Question 1 (20 points)

The following figure shows an example of Monte Carlo Tree Search, where each leaf node is annotated with the utility value. To maximize the utility, here Monte Carlo Tree Search is applied to return a planning path. Suppose that after several steps, the current state of the MCTS search tree is also shown in the following figure, where the numbers within a node represent "estimated value / the number of visits" and those nodes encircled by dashed lines are not expanded yet.

- ① (5 points) Provided that the hyperparameter $C=1$ in the UCB bound, please show the path selected by MCTS.
- ② (5 points) Please continue the procedures of selection, expansion, roll out, and backpropagation, and visualize the state of the search tree afterwards.
- ③ (10 points) Try more iterations of the four procedures of MCTS, and see if the leaf node with the largest utility (i.e., 9) can be returned. If not, please give your suggestions on how to modify the algorithm for higher efficiency.

② a. from ① we select:

$$29/7 \rightarrow 21/4 \rightarrow 7/1$$

b. Expansion:

7/1 is a leaf node, no expansion needed

c. Rollout: utility = 7

d. back propagation:

$$1) 7/1 \Rightarrow (7+7)/(1+1) = 14/2$$

$$2) 21/4 \Rightarrow (21+7)/(4+1) = 28/5$$

$$3) 29/7 \Rightarrow (29+7)/(7+1) = 36/8$$

① a. From root, calculate UCB for 3 children: ($C=1$)

$$\text{left: } UCB_L = \frac{Q_L}{N_L} + \sqrt{\frac{\ln N_{root}}{N_L}} = 1 + \sqrt{\ln 7} = 1 + 1.395 = 2.395$$

$$\text{middle: } UCB_M = \frac{Q_M}{N_M} + \sqrt{\frac{\ln N_{root}}{N_M}} = \frac{21}{4} + \sqrt{\frac{\ln 7}{4}} = 5.25 + 0.697 = 5.947 \text{ [chosen]}$$

$$\text{right: } UCB_R = \frac{Q_R}{N_R} + \sqrt{\frac{\ln N_{root}}{N_R}} = \frac{7}{2} + \sqrt{\frac{\ln 7}{2}} = 3.5 + 0.986 = 4.486$$

b. calculate third layer:

$$\text{left: } UCB'_L = \frac{Q_L}{N_L} + \sqrt{\frac{\ln N_{parent}}{N_L}} = 6 + \sqrt{\frac{\ln 4}{1}} = 6 + 1.177 = 7.177$$

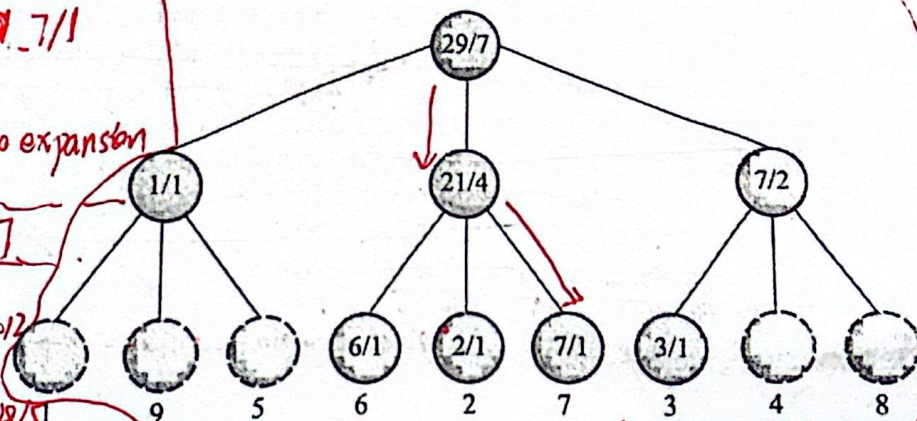
$$\text{middle: } UCB'_M = \frac{Q_M}{N_M} + \sqrt{\frac{\ln N_{parent}}{N_M}} = 2 + \sqrt{\frac{\ln 4}{1}} = 3.177$$

$$\text{right: } UCB'_R = \frac{Q_R}{N_R} + \sqrt{\frac{\ln N_{parent}}{N_R}} = 7 + 1.177 = 8.177 \text{ [chosen]}$$

Conclusion: Selected Path:

$$(29/7 \rightarrow 21/4 \rightarrow 7/1)$$

In using code to test ③



③ $C < 6$, fail to find the "utility 9" node with root iteration (brute force)

Suggestions:

1) Increase C to 6

2) Consider ϵ -greedy

This conclusion was tested by code written by ~~code~~



Question 2 (20 points)

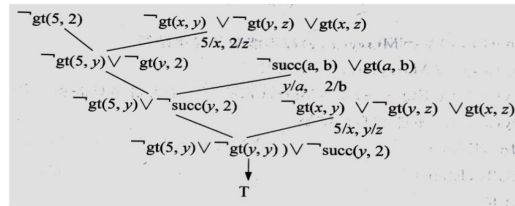
Q2:

Suppose that you are given the following facts.

- $\forall x \forall y \forall z \text{ gt}(x, y) \wedge \text{gt}(y, z) \rightarrow \text{gt}(x, z)$
- $\forall a \forall b \text{ succ}(a, b) \rightarrow \text{gt}(a, b)$
- $\forall x \text{ gt}(x, x)$

To prove $\text{gt}(5, 2)$ with resolution, there is an attempt shown as below.

- $\forall x \text{ gt}(x, x)$



- What kind of errors exist(s) in the above proof?
- To avoid such errors, do you need additional procedures? What kind of procedures? Can you re-write the correct proof?

(1) Wrong Substitution: $5/x, 2/z$ should be $x/5, z/2$

(2) Invalid substitution ~~z~~ $z \rightarrow y$ in $\neg \text{gt}(y, z)$.

Assume $y, z = 2, 1$, $\text{gt}(2, 1)$ ~~and~~ not equals $\text{gt}(2, 2)$.

(3) Because of (2) is wrong, the following final inference is wrong too, so can not get the final "T".

(b) Need Additional Procedures:

(I) $\text{succ}(3, 2)$

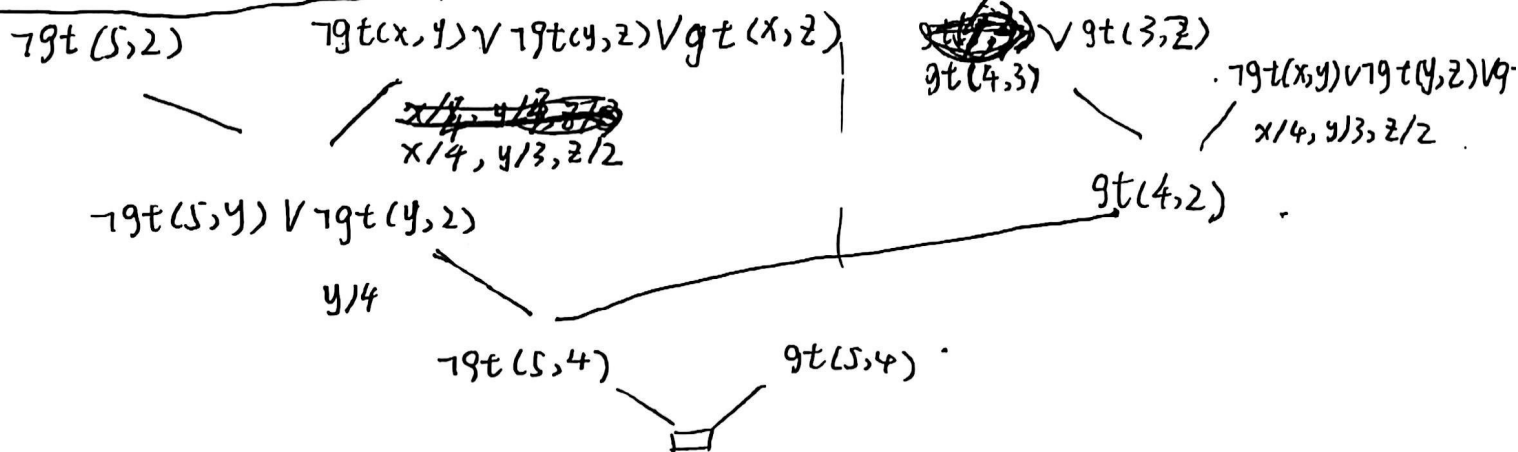
(II) $\text{succ}(4, 3)$

(III) $\text{succ}(5, 4)$

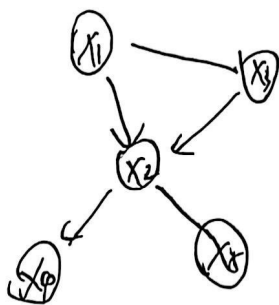
Using Transitivity chain

$\left. \begin{array}{l} \text{gt}(3, 2) \\ \text{gt}(4, 3) \\ \text{gt}(5, 4) \end{array} \right\}$

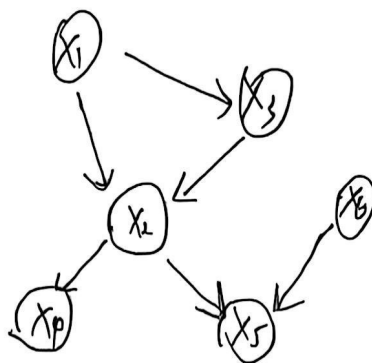
so proof could be:



Question 3:



left



right

For left: $P(X_1, X_2, X_3, X_4, X_5) = P(X_1) P(X_3|X_1) P(X_2|X_1, X_3) P(X_4|X_2) P(X_5|X_2)$.

(1) $P(X_3|X_4)$, order of elimination: X_5, X_1, X_2

$$P(X_3|X_4) = \sum_{X_5} \sum_{X_1} \sum_{X_2} P(X_1) P(X_3|X_1) P(X_2|X_1, X_3) P(X_4|X_2) P(X_5|X_2) \quad (1-1)$$

① Eliminate X_5 , involving $P(X_5|X_2)$

$$f_1(X_2) = \sum_{X_5} P(X_5|X_2) = 1 \quad (\text{Total probability}) \quad (1-2)$$

② Eliminate X_1 , involving $P(X_1) P(X_3|X_1) P(X_2|X_1, X_3)$,

$$f_2(X_2, X_3) = \sum_{X_1} P(X_1) P(X_3|X_1) P(X_2|X_1, X_3) = \sum_{X_1} P(X_1, X_3) P(X_2|X_1, X_3) \quad (1-3)$$

③ Eliminate X_2 , involving X_2

$$\text{Current Remaining: } P(X_3, X_4) = \sum_{X_2} f_2(X_2, X_3) P(X_4|X_2) = f_3(X_3, X_4) \quad (1-4)$$

$$\text{Final: } P(X_3|X_4) = \frac{P(X_3, X_4)}{P(X_4)} = \frac{f_3(X_3, X_4)}{\sum_{X_3} f_3(X_3, X_4)}$$

$$\text{Max (factor size)} = \max[f_1(X_2), f_2(X_2, X_3), f_3(X_3, X_4)] = 2^2 = 4$$

(2) $P(X_3|X_4)$, order of elimination: X_2, X_1, X_5

① Eliminate X_2 , involving $P(X_2|X_1, X_3), P(X_4|X_2), P(X_5|X_2)$

$$f_1(X_1, X_3, X_4, X_5) = \sum_{X_2} P(X_2|X_1, X_3) P(X_4|X_2) P(X_5|X_2)$$

$$\text{Current: } P(X_3|X_4) = \sum_{X_1} \sum_{X_5} P(X_1) P(X_3|X_1) f_1(X_1, X_3, X_4, X_5)$$

② Eliminate X_1 :

$$f_2(X_3, X_4, X_5) = \sum_{X_1} P(X_1) P(X_3|X_1) f_1(X_1, X_3, X_4, X_5)$$

③ Eliminate X_5 :

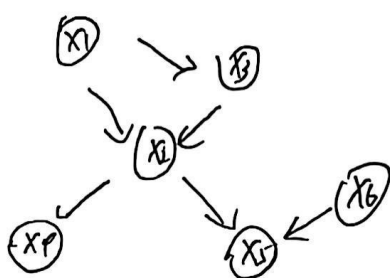
$$f_3(X_3, X_4) = \sum_{X_5} f_2(X_3, X_4, X_5) = P(X_3, X_4)$$

$$\text{Final: } P(X_3|X_4) = \frac{f_3(X_3, X_4)}{\sum_{X_3} f_3(X_3, X_4)}$$

$$\text{Max Factor size} = \max(f_1(X_1, X_3, X_4, X_5)) = 2^4 = 16, \text{ much larger than } 2^2 \text{ in (1)}$$



Right:



(1) 5126

$$P(X_1, X_2, X_3, X_4, X_5, X_6) = P(X_1) P(X_3 | X_1) P(X_2 | X_1, X_3) P(X_4 | X_2) P(X_6) P(X_5 | X_2, X_3, X_6)$$

① Eliminate X_5 , involving $P(X_5 | X_2, X_3)$

$$f_1(X_2, X_3) = \sum_{X_5} P(X_5 | X_2, X_3) = 1 \quad (\text{Total probability})$$

② Eliminate X_1 , involving $P(X_1) P(X_3 | X_1) P(X_2 | X_1, X_3)$

$$f_2(X_2, X_3) = \sum_{X_1} P(X_1) P(X_3 | X_1) P(X_2 | X_1, X_3)$$

③ Eliminate X_2 , involving $f_2(X_2, X_3), P(X_4 | X_2)$

$$f_3(X_3, X_4) = \sum_{X_2} f_2(X_2, X_3) P(X_4 | X_2)$$

④ Eliminate X_6 , involving $P(X_6) = 1$

$$\text{Final } P(X_3, X_4) = f_3(X_3, X_4) / \sum_{X_3} f_3(X_3, X_4)$$

$$\text{Max Factor size} = O[f_3(X_3, X_4)] = f_2(X_2, X_3) = 2^2 = 4$$

2) 2156

① Eliminate X_2 , involving $P(X_2 | X_1, X_3) P(X_4 | X_2) P(X_5 | X_2, X_6)$

$$f_1(X_1, X_3, X_4, X_5, X_6) = \sum_{X_2} P(X_2 | X_1, X_3) P(X_4 | X_2) P(X_5 | X_2, X_6)$$

② Eliminate X_1 , involving ~~f_1~~ $P(X_1) P(X_3 | X_1)$

$$f_2(X_3, X_4, X_5, X_6) = \sum_{X_1} P(X_1) P(X_3 | X_1) f_1(X_1, X_3, X_4, X_5, X_6)$$

③ Eliminate X_5 , involving f_2

$$f_3(X_3, X_4, X_6) = \sum_{X_5} f_2(X_3, X_4, X_5, X_6)$$

④ Eliminate X_6 , involving $P(X_6)$

$$f_4(X_3, X_4) = \sum_{X_6} f_3(X_3, X_4, X_6) P(X_6), \quad \sum_{X_6} P(X_6) = 1 = f_3(X_3, X_4, X_6)$$

$$\text{Final: } P(X_3, X_4) = \frac{f_4(X_3, X_4)}{P(X_4)}$$

$$\text{Max Factor size} = O[f_1(X_1, X_3, X_4, X_5, X_6)] = 2^6 = 32 > 4 \text{ in (1)}$$

